AMERICAN ASSOCIATION of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL

Transforming Technology, Modern Practice, and Clinical Impact

JUNE 16–20 | UNIVERSITY OF MICHIGAN

Biological, Molecular, and Functional-Guided ART

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Disclosure

- None

Outline

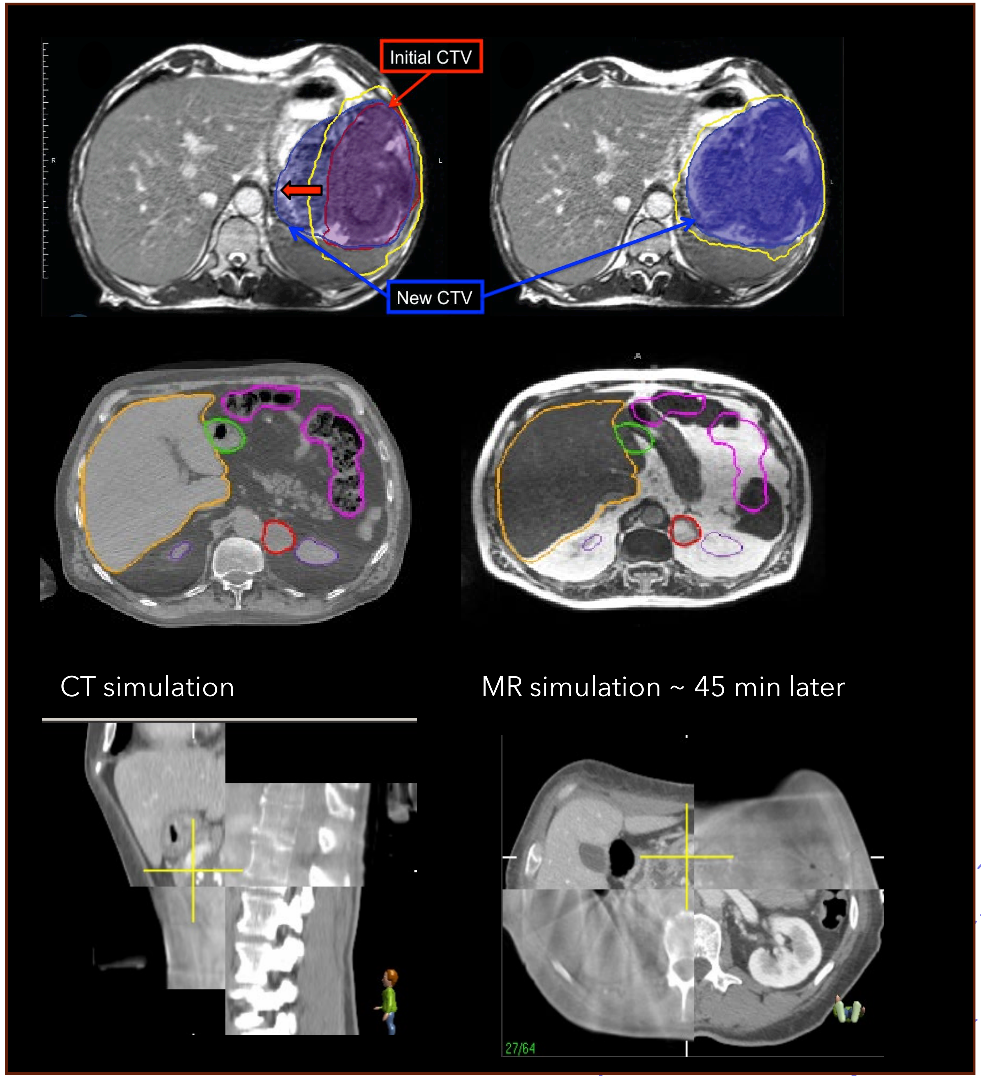
- Rational for biological guided ART
- Integration of biological guided ART
- Clinical examples of biological guided ART and strategies
- Challenges

What type of ARTs are used in your center?

- Anatomy based ART
- Anatomy and Biology based ART
- IGRT only No ART

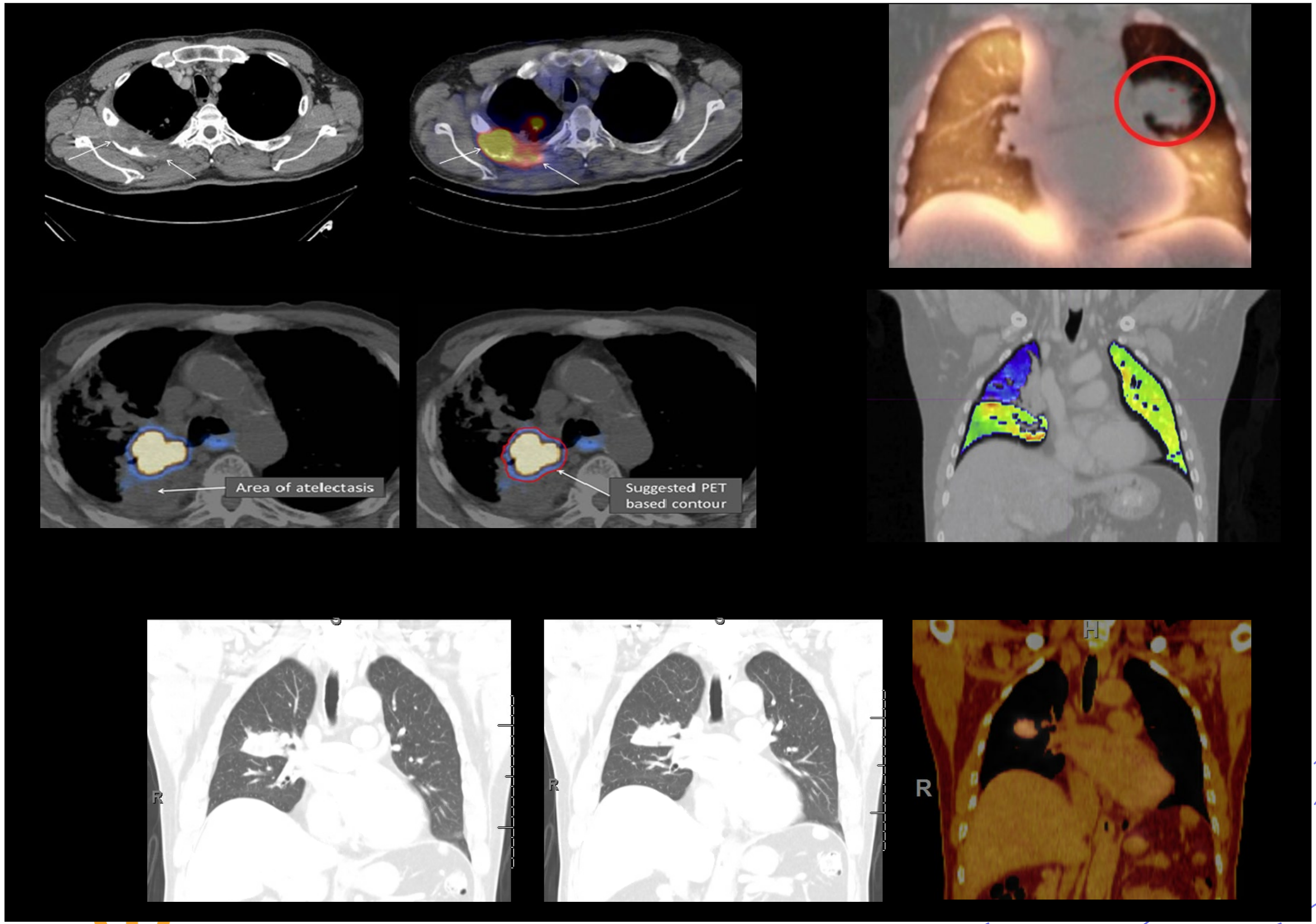
Why Anatomy-based Adaptive Radiotherapy (ART) Is Not Enough?

- Anatomy based ART accounts for
  - Tumor/OAR size change
  - Tumor/OAR shape change
  - Positional change
  - Patient body difference



Why Anatomy-based Adaptive Radiotherapy (ART) Is Not Enough?

- Limitations
  - Not reflecting tumor/tissue biology/physiology
  - OAR functionally heterogeneous
  - Volume reduction is often a late response



Konert et al. Radiotherapy and Oncology 2015
Elojeimy et al. AJR2016
Vinogradskiy et al. IJROBP 2018.

Definitions

- Biology guided ART is an advanced form of adaptive radiotherapy that **dynamically** modifies treatment based not only on **anatomical** changes, but also on the evolving **biological** and **physiological** characteristics of the tumor and surrounding normal tissues.

Key features

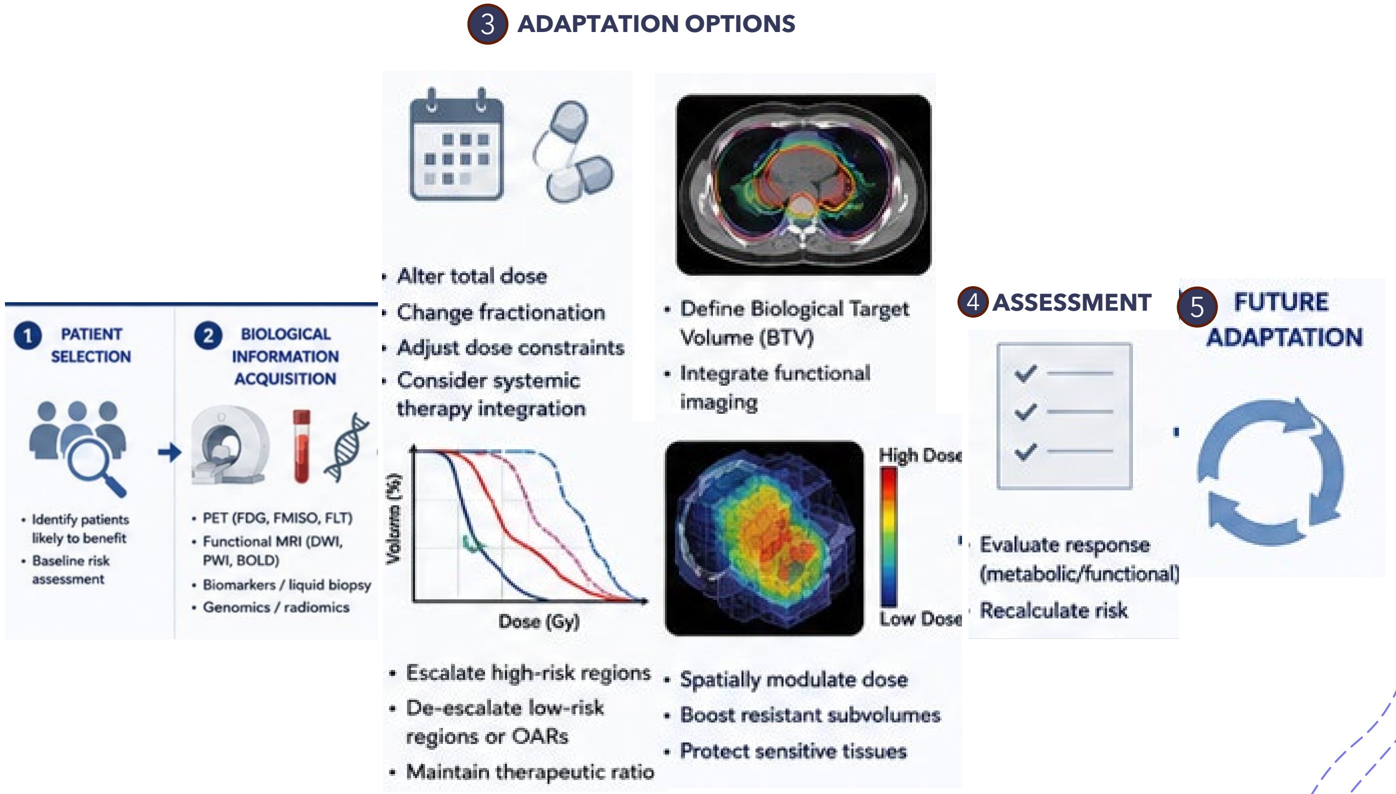
- Changes beyond anatomy: metabolic activity, hypoxia, proliferation, radiosensitivity, immune activation, tumor microenvironment dynamics, .etc.
- Detections: Functional CT (ventilation CT), Functional MRI (DWI, BoldMRI), PET (FDG, FMISO, PSMA, etc.), SPECT, Liquid Biopsy, .etc.
- Dose re-optimization: dose escalation, de-escalation, dose sparing.
- Objectives: Overcome radio-resistance, preserve functional normal tissues, immune response.

Integrating Biological Information into ART Workflows

- Timing of functional adaptation
  - Prior to anatomic changes vs. concurrent with anatomic changes
  - Between fractions (offline) vs. during fractions (online)
- Scale of Functional Adaptation
  - Patient scale: biomarkers for risk stratification
  - Region scale: targeting high-risk/residual disease
  - Voxel scale: spatial response maps
- Tumor vs. Normal Tissue Functional Adaptation
  - Modulating therapeutic window / ratio

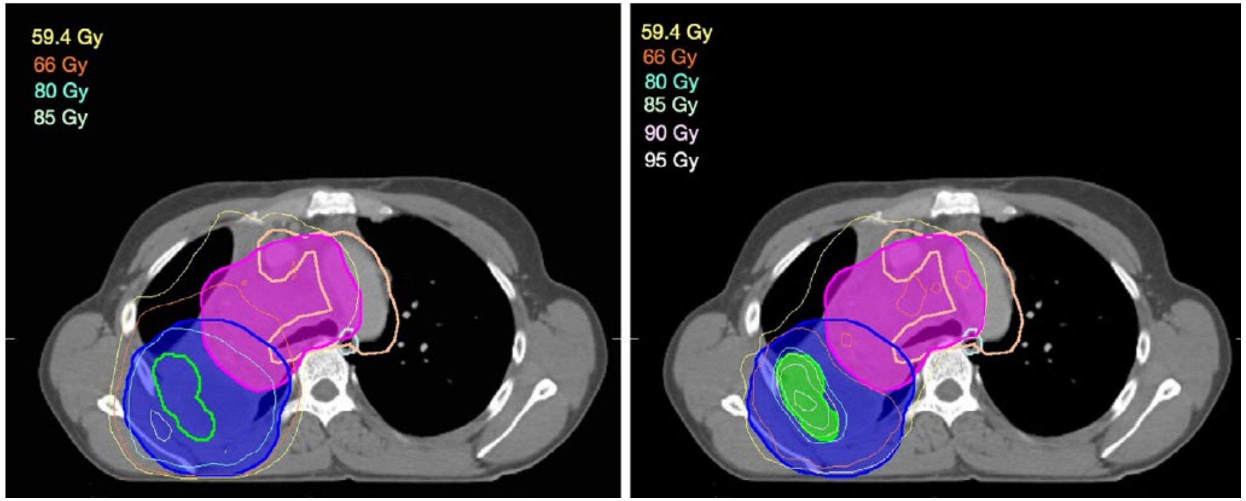
Integrating Biological Information into ART Workflows

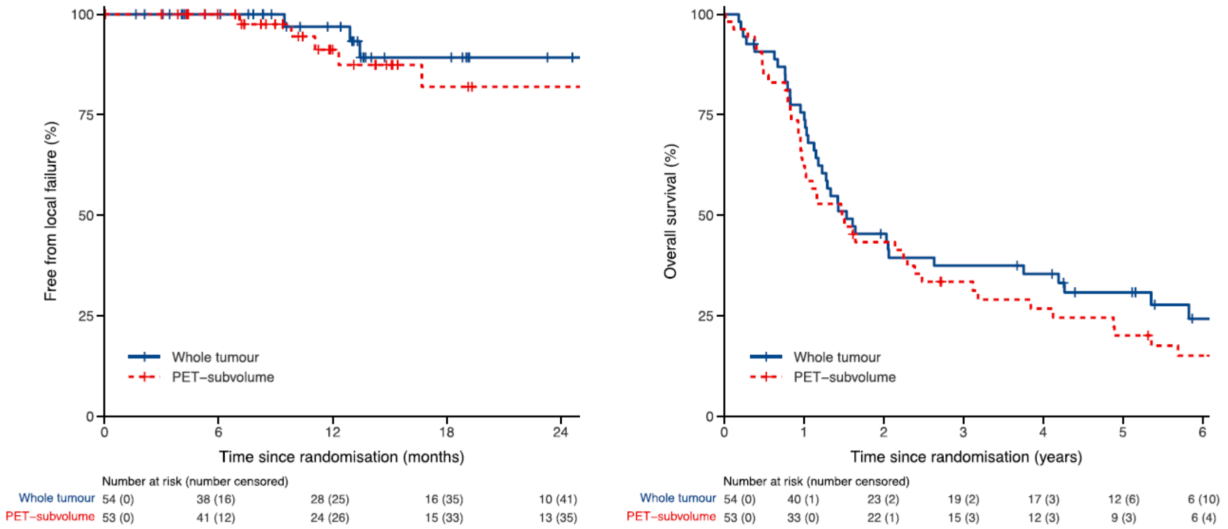
- Biological information incorporation/utilization
  - Patient selection
  - Dose regime change
  - Delineation
  - Dose escalation/de-escalation
  - Dose painting
- Assessment and re-adaptation
  - Follow up scans
  - Future adaptation



LA-NSCLC Boosting

- Entire primary tumor vs only high risk PET-defined sub-volume (FDG).
- Dose escalation achieved as much as possible (>85Gy in 24 fractions and limited by OAR constraints)
- Similar local control.

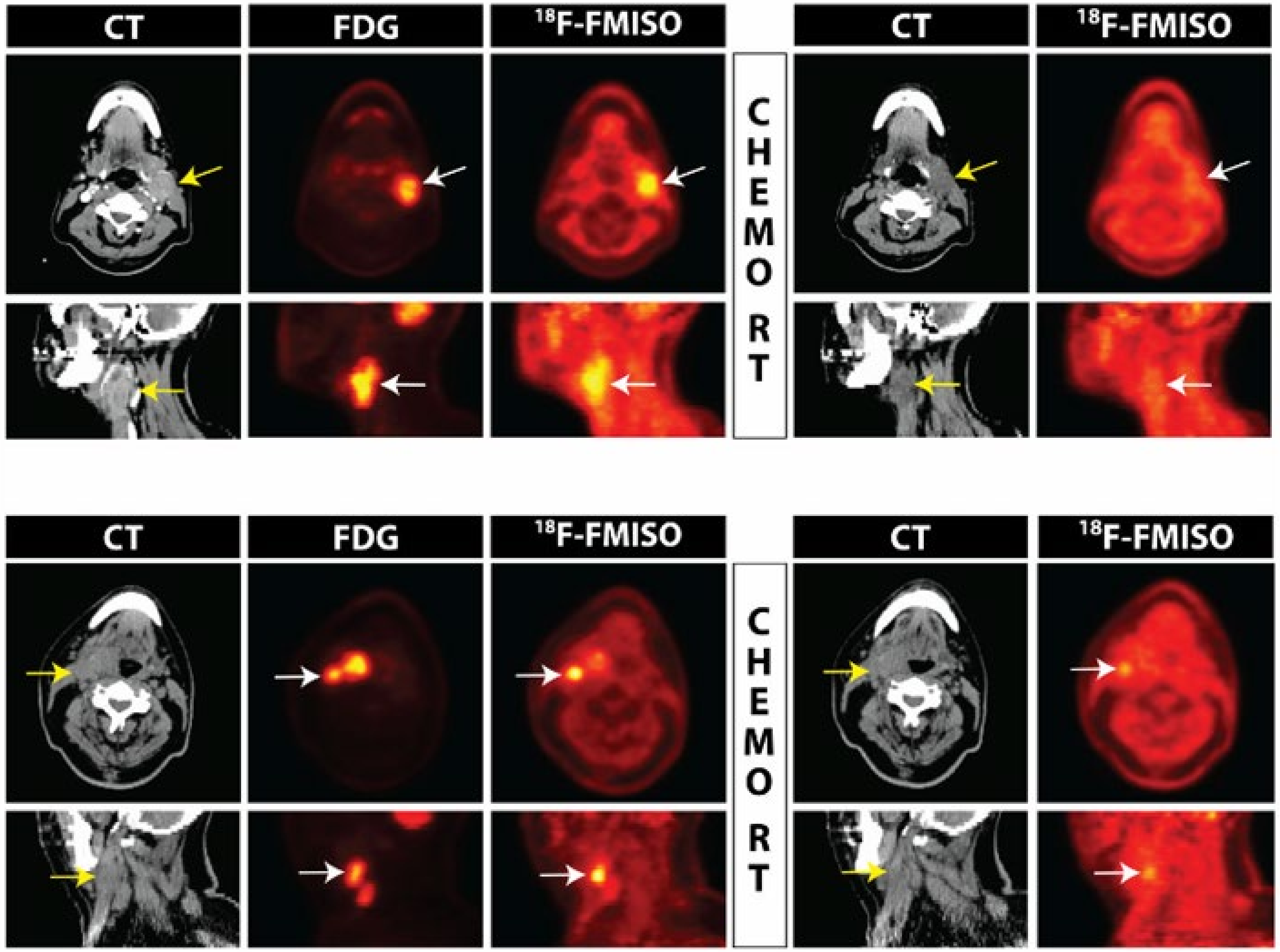




Cooke et al. Radiotherapy and Oncology 2023

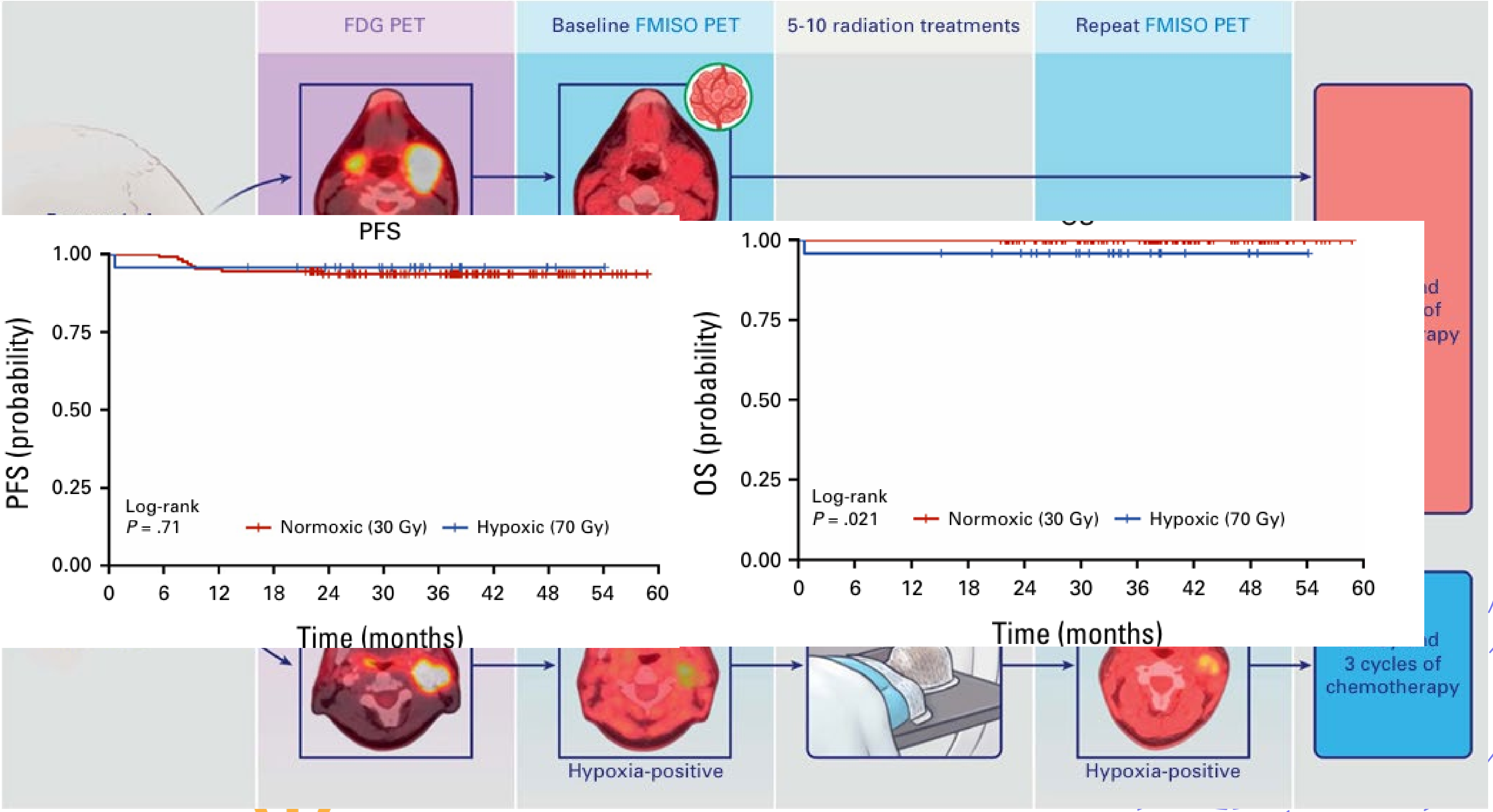
Hypoxia-Directed HN Treatment (dose de-escalation)

- HPV-related Oropharyngeal cancer
- FDG PET and FMISO PET taken at pretreatment and one week into treatment.
- Dose regime is adapted based on Hypoxia.



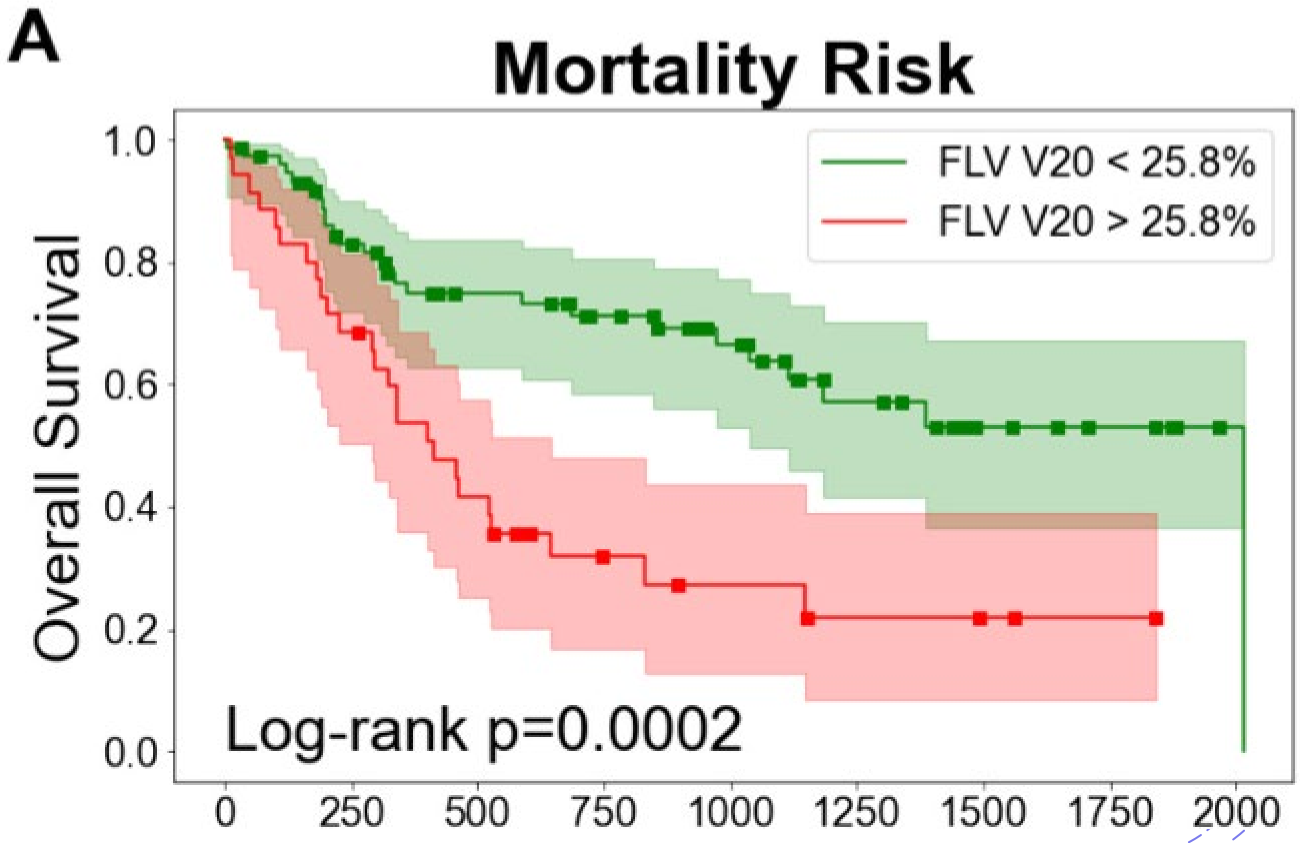
Hypoxia-Directed HN Treatment

- HPV-related Oropharyngeal cancer
- 70Gy vs 30Gy



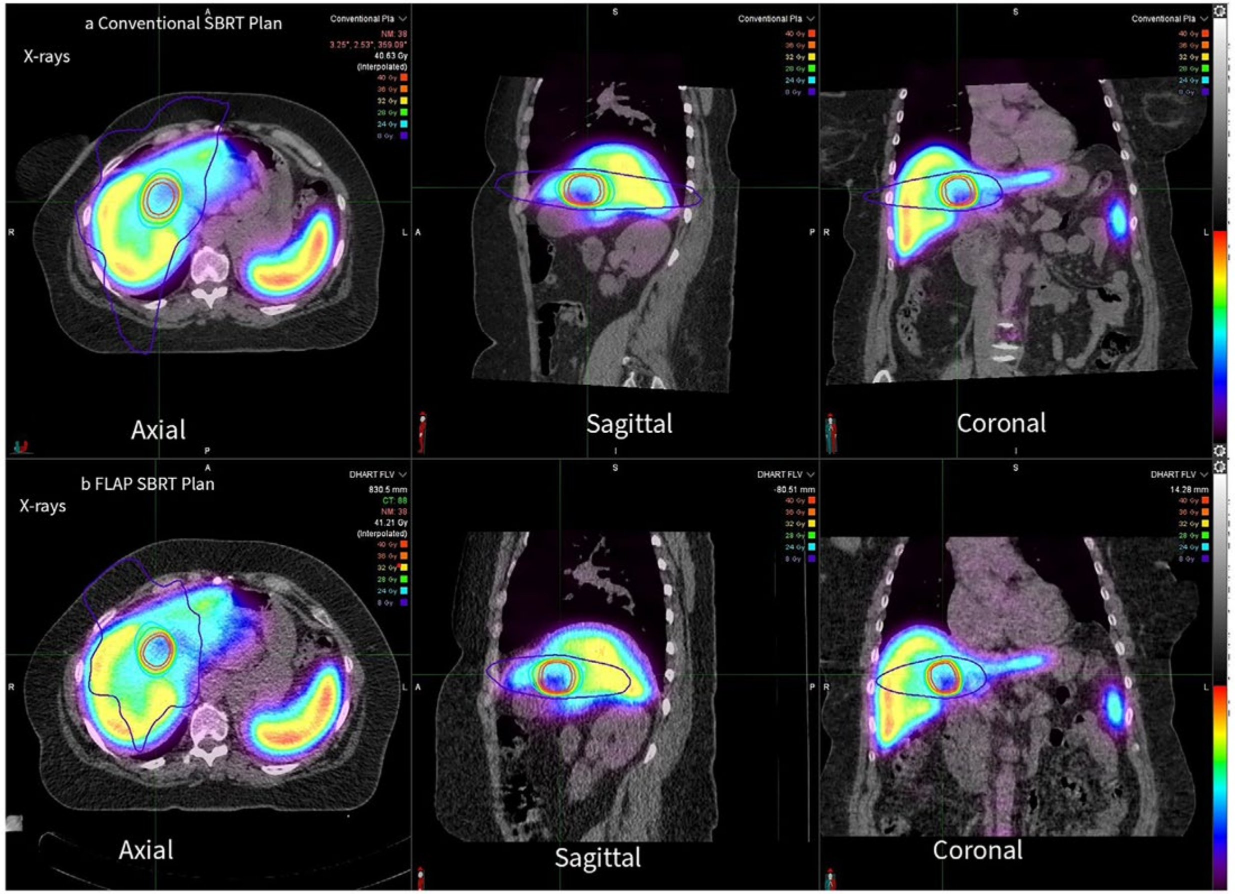
Functional liver avoidance planning (FLAP)

- 99mTc Sulfur Colloid SPECT/CT imaging determine functional liver volume (FLV).
- Liver function dose-response curve modeling.
- FLV dosimetry constraints determination.



Functional liver avoidance planning (FLAP)

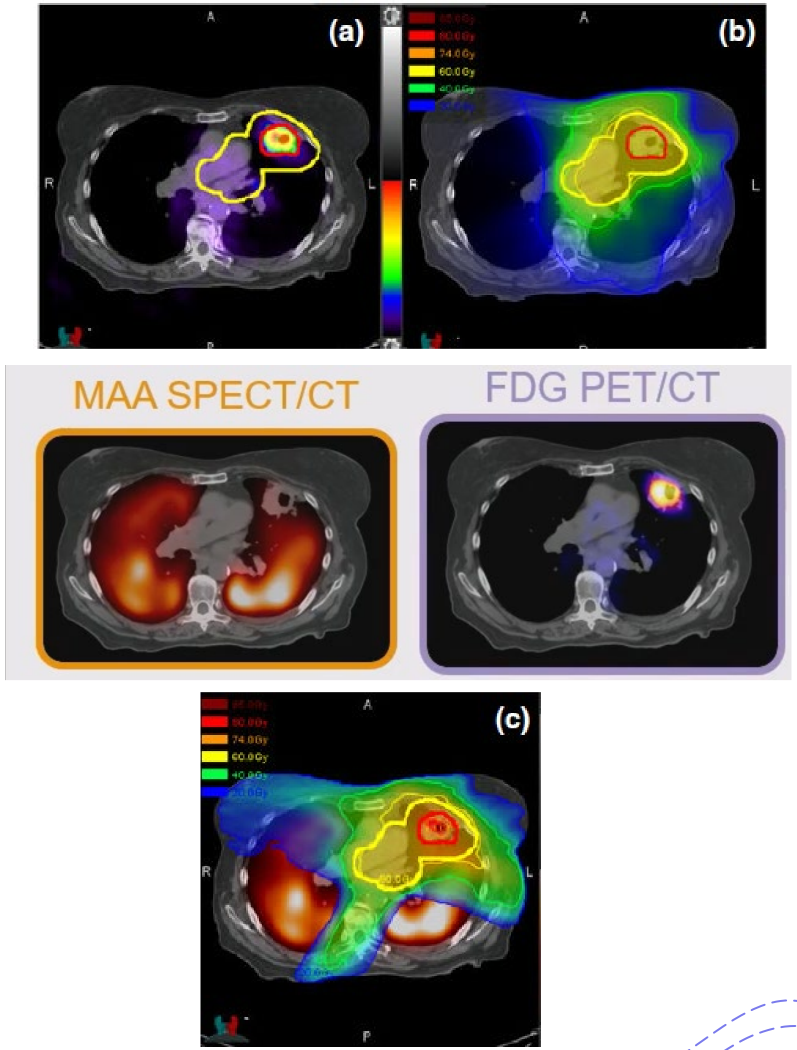
- Plan optimization considering FLV avoidance
- FLAP V16Gy<12%, FLAP Dmean<1000cGy



Schaub et al. IJROBP 2018
Tsai et al. radiotherapy and oncology 2025
Yorkeetal et al. PRO 2026

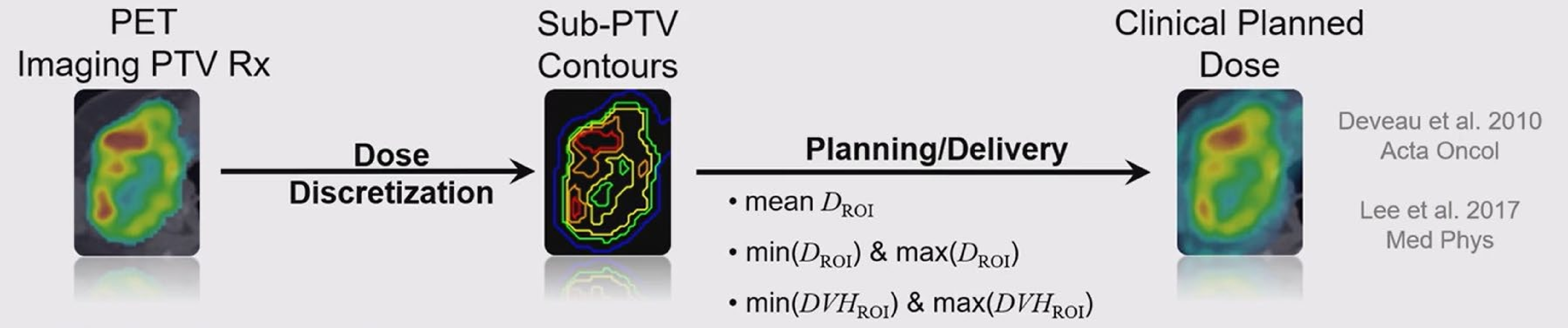
Functional Lung avoidance and response-adaptive escalation (FLARE)

- MAA SPECT/CT for lung function determination + FDG PET/CT for tumor metabolism.
- Plan reoptimized based on dose escalation and OAR sparing



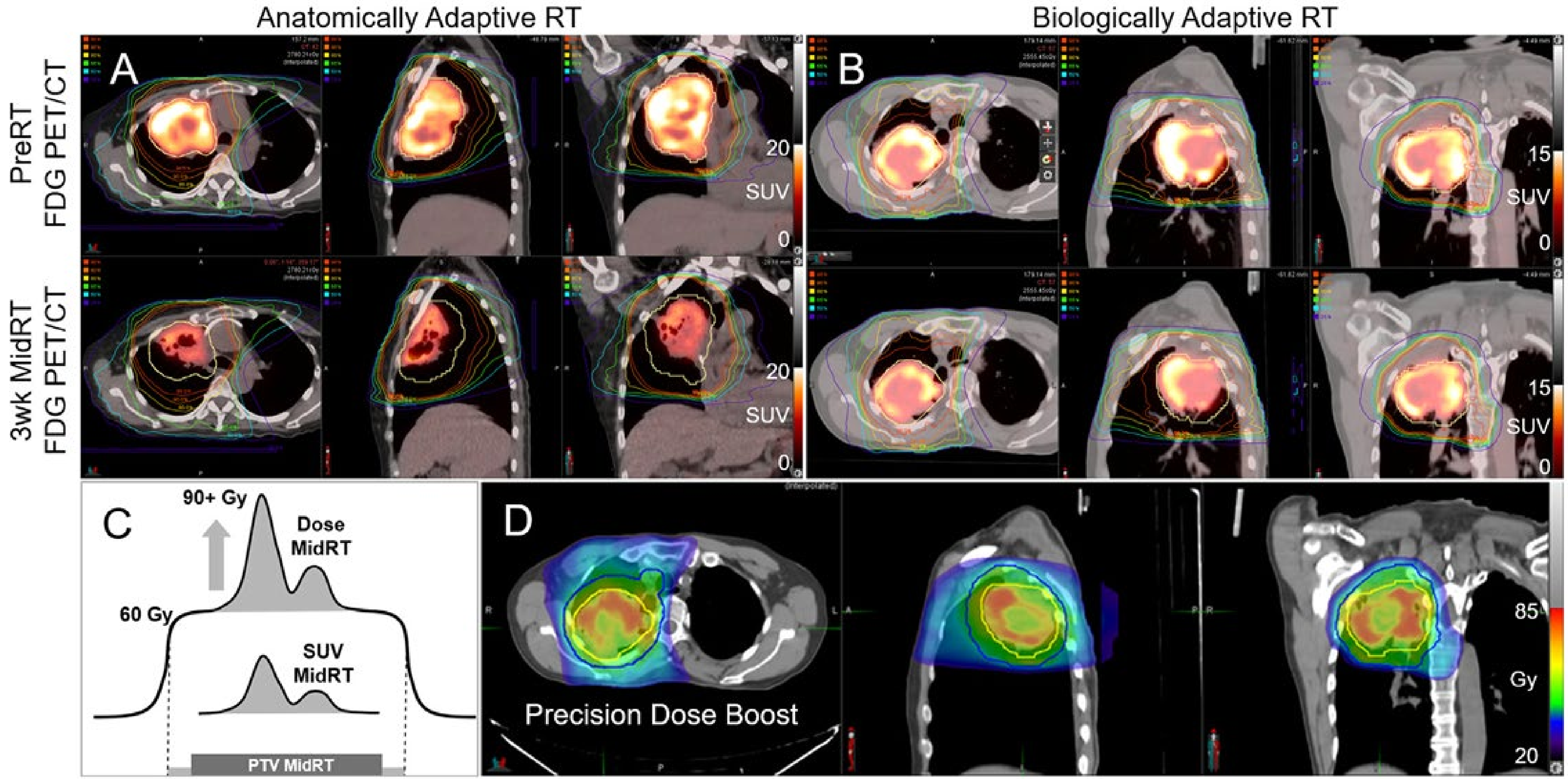
Lee et al. MedPhys 2017

Functional Lung avoidance and response-adaptive escalation (FLARE)



Lee et al. MedPhys 2017

Functional Lung avoidance and response-adaptive escalation (FLARE)



Zeng et al. Seminars in Radiation Oncology 2020

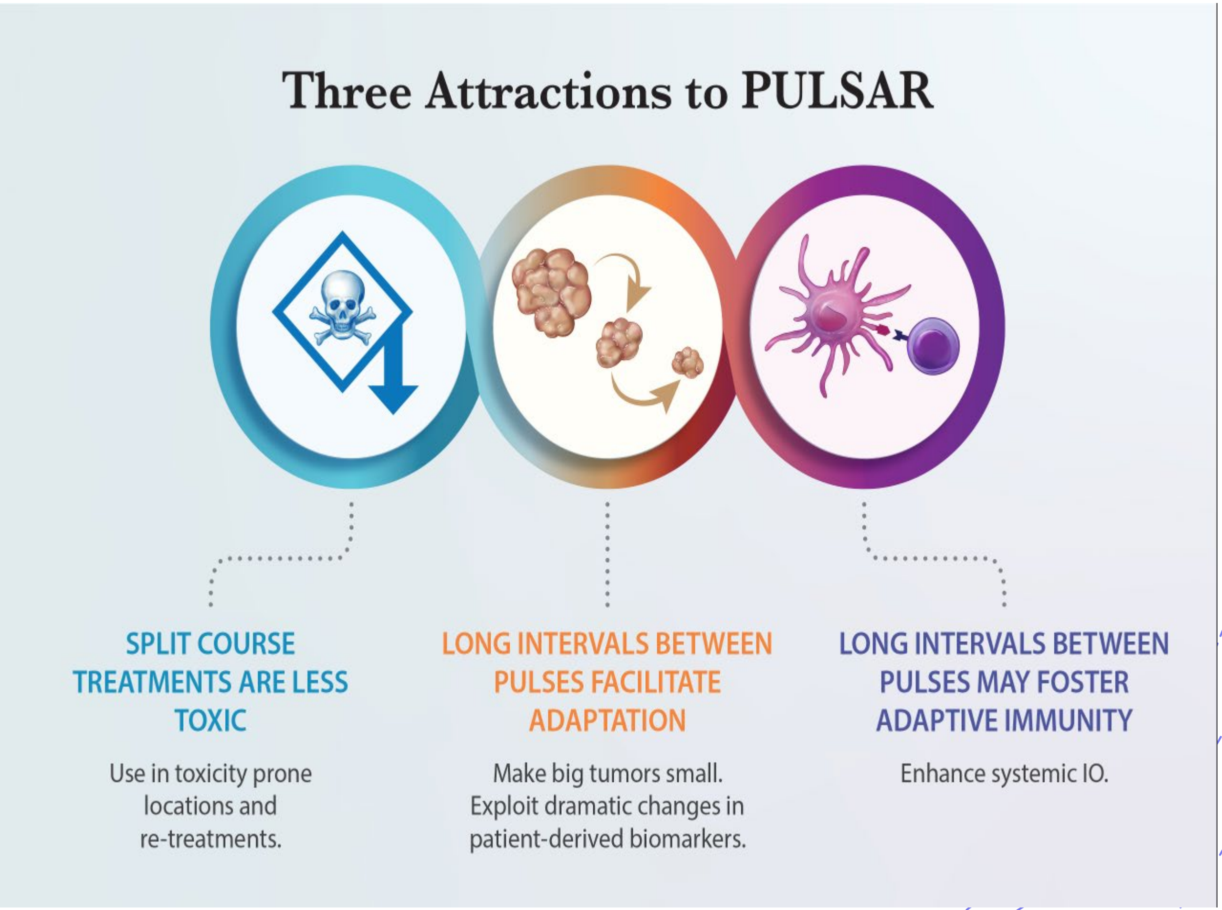
Pulsar

- PULSAR – Personalized ULtra-fractionated Stereotactic Adaptive Radiotherapy
  - Step 1: Use a high dose "pulse" of focused radiation to halt tumor progression

    -Dose high enough to afford months of durable control
  - Step 2: STOP (for at least a week, typically a month) -- Monitor **biological** processes that follow and Interrogate both the patient and the patient's tumor
  - Step 3: Plan the next pulse tailored to the patient -- Based on sophisticated **imaging** and acquired **biological** information – ideally in combination with the ideal systemic therapy partner
- More than one pulse can be given
- Until interrogation measures cure (or failure of the approach)

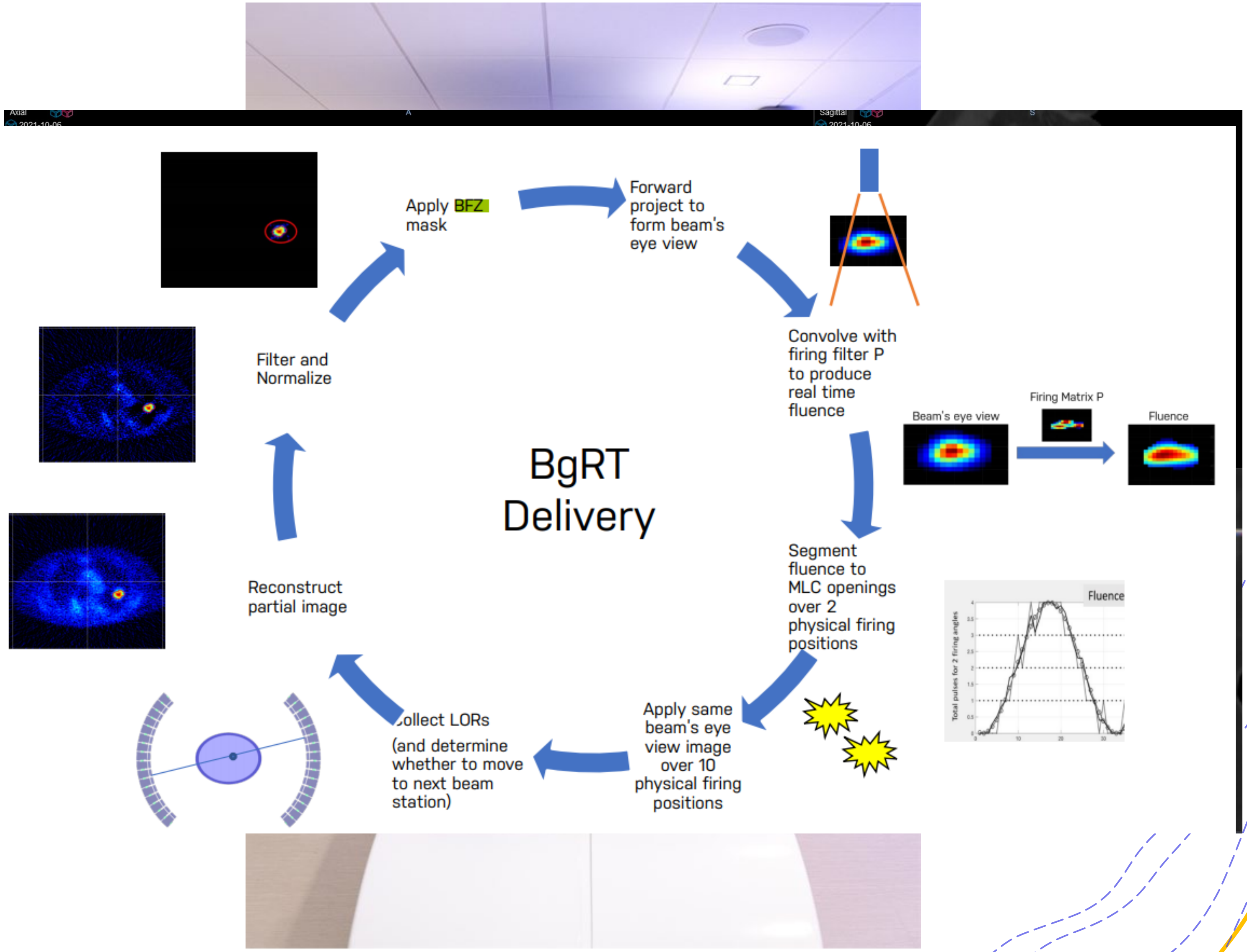
Pulsar

- Split course treatments are less toxic
  - For the same SAbR dosing, fractions are more toxic than pulses
  - Predicted by historical split course trials (all were less toxic)
  - Use for tumors in toxicity prone locations
- Long intervals between pulses facilitate adaptation
  - Time allows for change (e.g., tumor shrinkage)
  - Change facilitates meaningful adaptation (e.g., re-planning)
  - Use for big tumors
  - This can be part of true personalization of Rx
- Long intervals between pulses foster adaptive immunity
  - Complete the cascades required for a vaccine
  - Use in combo with systemic IO

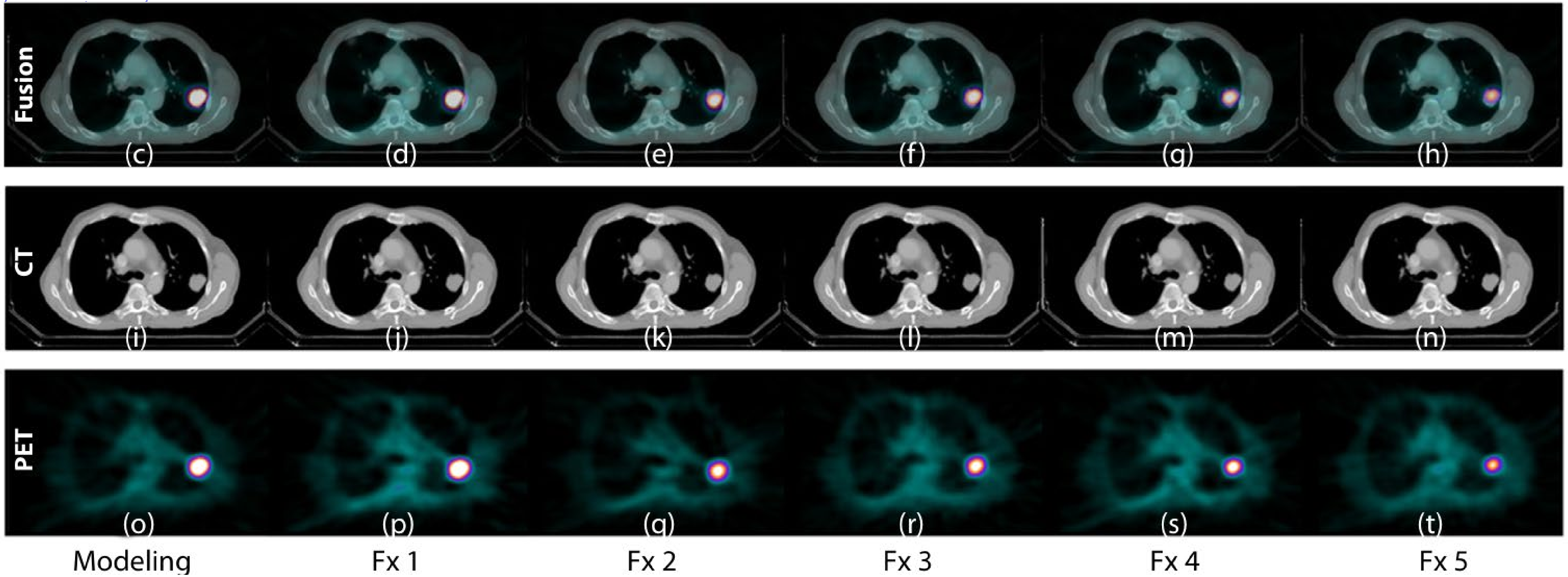


Potential online BGRT

- RefleXion PET-linac
- Real time target tracking and dose delivery based on PET signal.
- Approved for Bone and Lung by FDA.



PET-linac based ART



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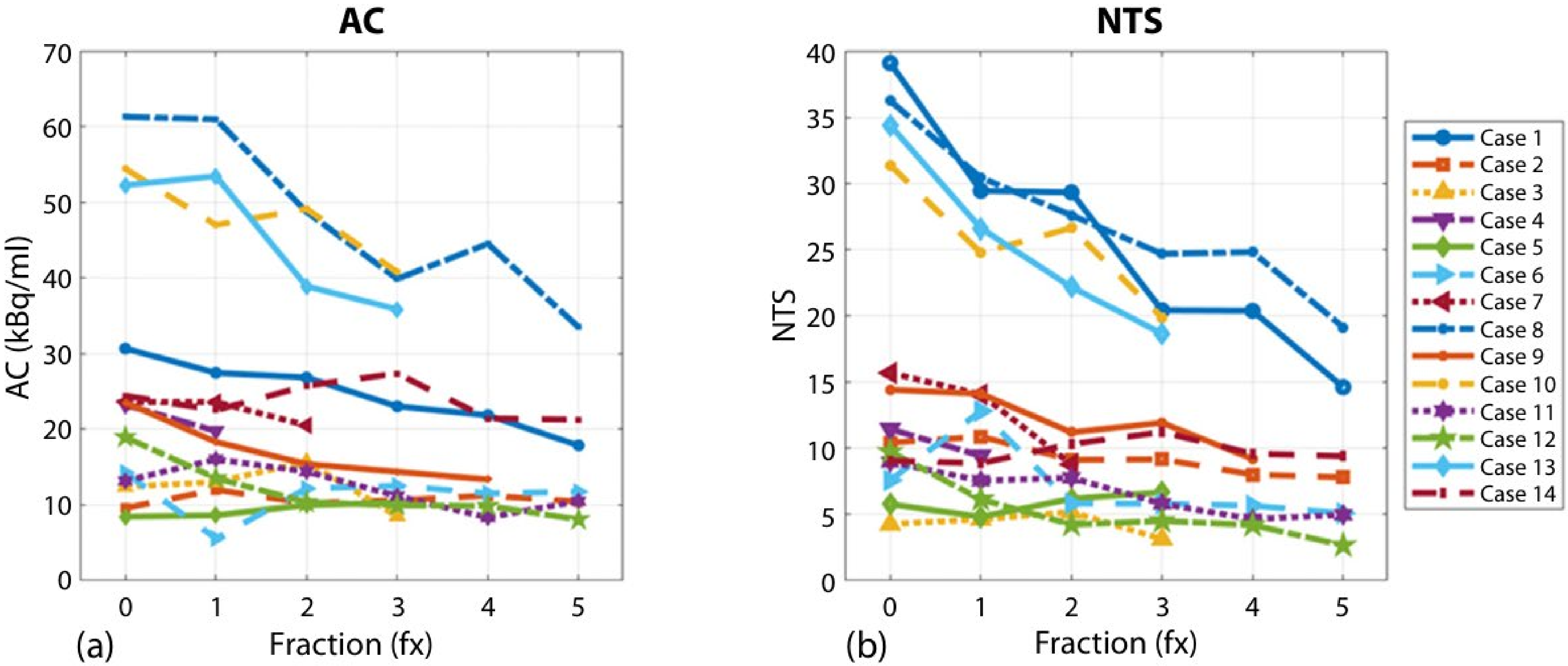
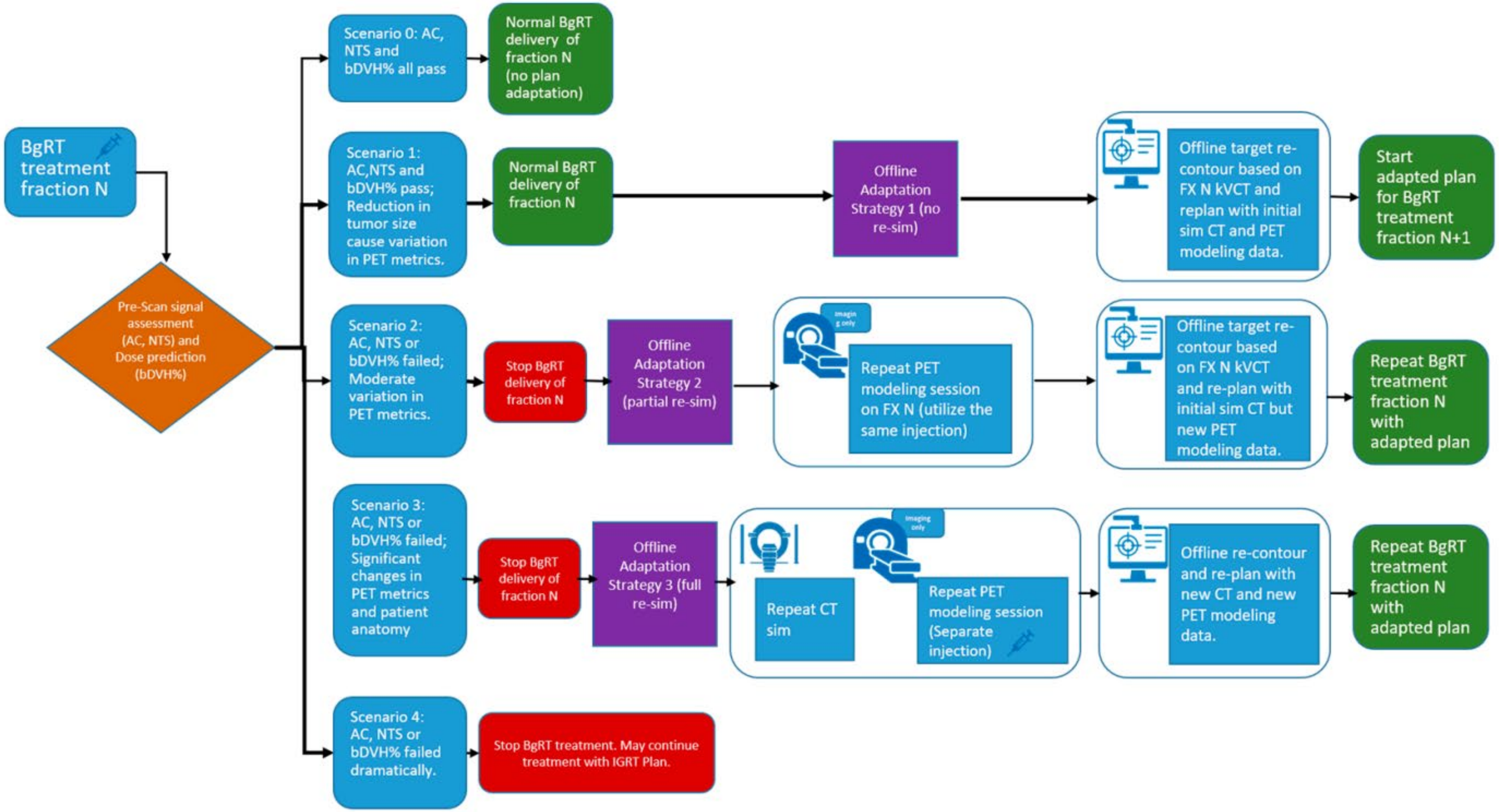
Lin et al. IJROBP 2025

PET-linac based ART



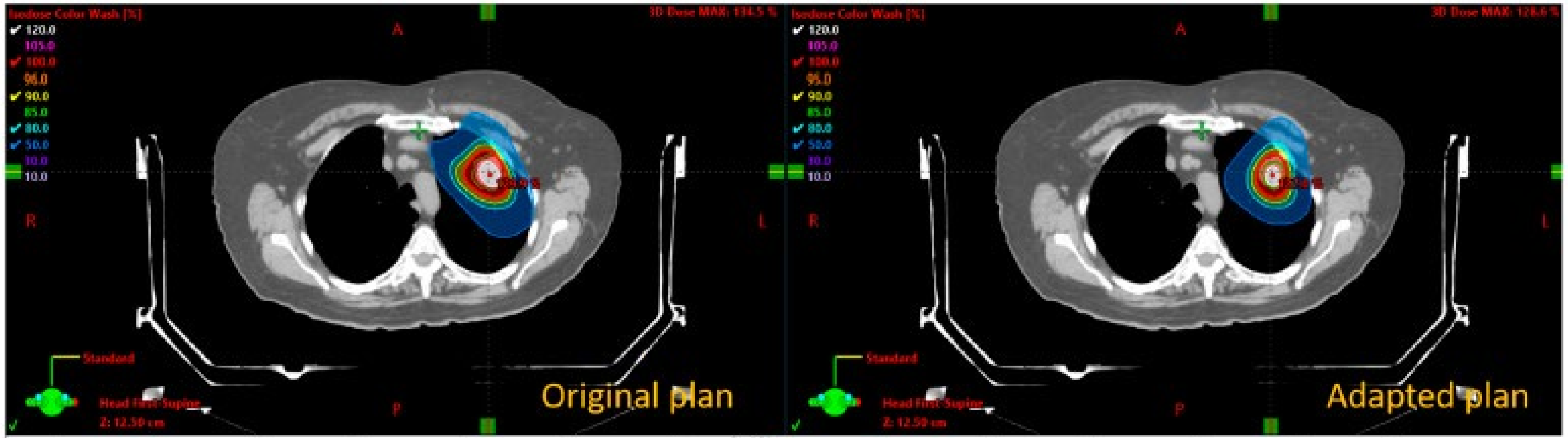
Lin et al. IJROBP 2025

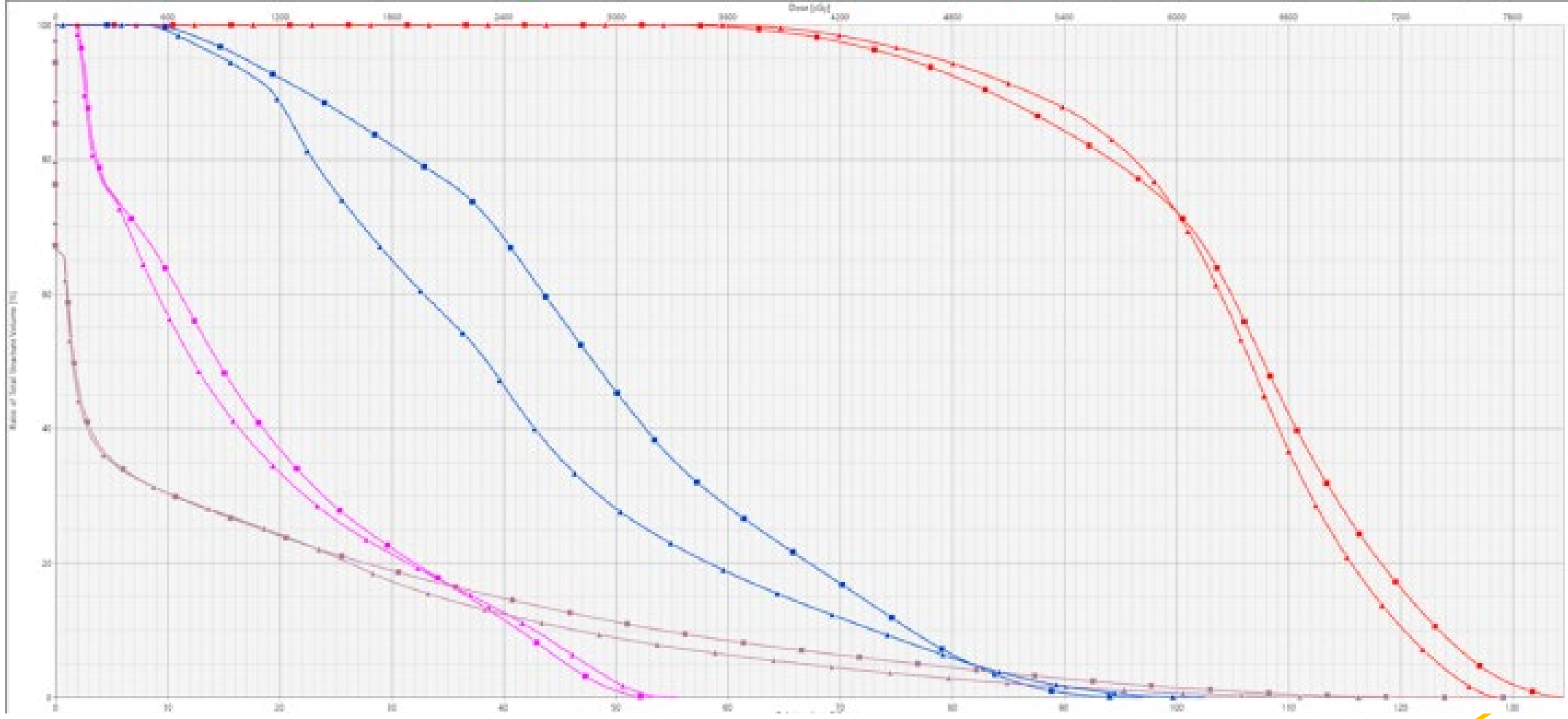
PET-linac based ART



Cai et al. Cancers 2025

PET-linac based ART





Cai et al. Cancers 2025

Challenges in biology guided adaptation

- Image or functional information acquisition burden.
- Optimal timing for adaptation.
- Technical uncertainty in functional imaging or information handling.
- Workflow and staffing complexity.
- Robust optimization considering both anatomical and functional information.
- Uncertain clinical benefit
- Regulatory, reimbursement, and implementation barriers

How interested are you in implementing biology- or function-guided adaptive radiotherapy after seeing this presentation?

- More interested
- Less interested
- Same.

Acknowledgment

- Mayo Clinic Florida: Jun Tan, Justin Park, Sridhar Yaddanapudi, Bo Lu, Chris Beltran, Laura Vallow
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- Vendor collaborators
  - Varian team and AIC
  - RefleXion team
  - Elekta team
  - United Imaging team
  - Sun Nuclear team
  - PTW team
- All collaborators over the years

Thank you for your attentions!

Questions?

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